

Comp1804 Report

30/03/2026

Word count: 2930 Words

Executive summary.

For Task 1, a machine learning model was developed to classify parliamentary motions into policy topics using TF-IDF text encoding and linear classifiers, including Logistic Regression and Support Vector Machines. The final model achieved an accuracy of approximately 80%, demonstrating strong performance, although it did not fully meet the client's target of 86%. The model performed well on clearly defined topics but struggled with overlapping categories.

For Task 2, a prototype model was built to classify motions as important or not important using manually labelled data. A Logistic Regression model achieved an accuracy of 60%, outperforming the baseline of 50%. While the model shows the ability to learn meaningful patterns, its performance is limited by the subjective nature of the labels and the small dataset size.

1. Data exploration and assessment.

The dataset used in this study contains **592 parliamentary motions**, each associated with a policy topic and several auxiliary features. The primary input feature for the machine learning tasks is the *motion_text*, while the target variable for Task 1 is *motion_topic*.

Initial exploration showed that the dataset includes a mixture of textual, categorical, numerical, and boolean features. The text length varies significantly, with an average length of approximately 669 characters and a maximum exceeding 17,000 characters, indicating high variability in document size. This supports the use of text-based encoding techniques capable of handling varying input lengths.

Missing value analysis revealed that the key features required for modelling (*motion_text* and *motion_topic*) contained no missing values. However, some auxiliary features such as *speaker* and *party* had missing entries, and *aux_numeric_feature* contained a notable number of missing values. Since these features were not essential for the classification tasks, they were excluded from the

modelling process.

The original dataset contained 13 distinct motion topics with an imbalanced distribution, where some classes had significantly more samples than others. However, in order to align with the client's requirements, the dataset was filtered to include only the six relevant policy areas: terrorism and security, welfare and benefits, local government, Scotland devolution, climate and environment, and education. This filtering ensured that the model focuses only on the required classification task and avoids noise from unrelated categories.

Overall, the dataset is suitable for machine learning, with sufficient size, clear target labels, and meaningful textual content. However, class imbalance and variability in text length were identified as key challenges that could impact model performance.

2. Data splitting and cleaning.

To prepare the dataset for machine learning, several preprocessing steps were applied, including data filtering, cleaning, and splitting.

First, the dataset was filtered to include only the six policy topics specified by the client. This step ensured that the classification task is aligned with the project requirements and removed irrelevant categories that could introduce noise and reduce model performance.

Next, data cleaning was performed by removing any rows with missing values in the key features, specifically *motion_text* and *motion_topic*. These features are essential for the classification task, and retaining incomplete entries could negatively affect model training. Since these columns contained no missing values after inspection, minimal cleaning was required. Other features with missing values, such as *speaker*, *party*, and *aux_numeric_feature*, were not used in the modelling process and therefore did not require imputation.

The dataset was then split into training and test sets using an 80:20 ratio. A stratified sampling approach was applied during the split to preserve the class distribution of the target variable across both sets. This ensures that each class is proportionally represented in both training and testing data, which is particularly important for multi-class classification problems.

The use of a fixed random state ensures reproducibility of the results, allowing the same data split to be obtained across multiple runs. This is essential for consistent evaluation and comparison of different models and configurations.

Overall, the data splitting and cleaning process was straightforward due to the relatively clean nature of the dataset. The main focus was on ensuring alignment with the task requirements and maintaining a representative distribution of classes in both training and testing sets.

3. Data encoding.

To transform the textual data into a format suitable for machine learning models, a Term Frequency–Inverse Document Frequency (TF-IDF) representation was used. TF-IDF is a widely used technique in natural language processing that converts text into numerical feature vectors by capturing the importance of words relative to the entire dataset.

The *motion_text* feature was encoded using a TF-IDF vectorizer with a maximum feature size of 8000. This limits the vocabulary to the most informative terms while reducing noise from less relevant or rare words. English stop words were removed during preprocessing to eliminate commonly used words that do not contribute meaningful information to the classification task.

In addition, both unigrams and bi-grams were used (*ngram_range* = (1,2)) to capture not only individual words but also meaningful word combinations such as “climate change” or “local councils”. This improves the model’s ability to understand context and enhances classification performance, particularly for policy-related text where phrases carry significant meaning.

The TF-IDF representation results in a high-dimensional sparse matrix, which is well-suited for linear models such as Logistic Regression and Support Vector Machines. These models are efficient in handling large feature spaces and are commonly used for text classification tasks.

Other features available in the dataset, such as entity indicators (e.g., *has_entity_date*, *has_entity_person*, *has_entity_event_or_org*), were considered but not included in the final model. While these features may provide additional contextual information, the primary predictive signal is contained within the textual data. Including these features could be explored as a potential improvement but was not necessary for building a strong baseline model.

Overall, TF-IDF encoding provides an effective and interpretable representation of the text data, enabling the models to capture key patterns and distinctions between different policy topics.

4. Task 1: topic classification.

4a. Model building.

To address the topic classification task, supervised machine learning models were developed to predict the policy topic of parliamentary motions based on their textual content.

As a baseline, **Logistic Regression** was selected due to its effectiveness and efficiency in high-dimensional text classification tasks. Logistic Regression is a linear model that performs well with sparse TF-IDF features and provides a strong benchmark for comparison. The model was trained using TF-IDF encoded text features, with a maximum of 8000 features and inclusion of both unigrams and bi-grams.

Following the baseline implementation, further improvements were explored using a **Support Vector Machine (SVM)** with a linear kernel (LinearSVC). SVMs are particularly well-suited for text classification due to their ability to handle high-dimensional feature spaces and maximise the margin between classes. This often leads to improved generalisation performance compared to simpler linear models.

Hyperparameter tuning was performed to improve model performance. In particular, the regularisation parameter (C) of the SVM model was adjusted to balance model complexity and generalisation. Increasing the value of C allows the model to fit the training data more closely, while lower values encourage simpler models with better generalisation.

The use of bi-grams in the TF-IDF representation was another key optimisation. This allowed the model to capture meaningful phrases such as “climate change” and “local councils”, which are important for distinguishing between policy topics. Empirically, this resulted in improved classification performance compared to using only single-word features.

The final model selected for evaluation was the optimised linear model trained on TF-IDF features with bi-grams, as it provided the best balance between performance, interpretability, and computational efficiency.

4b. Model evaluation.

The performance of the final model was evaluated using a classification report, confusion matrix, and overall accuracy. The model achieved an accuracy of approximately **80%** on the test set, indicating a strong ability to correctly classify parliamentary motions into their respective policy topics.

A detailed analysis of the classification report shows that the model performs well on several classes, particularly *Terrorism laws - For* and *Reduce Spending on Welfare Benefits*, which achieved high recall scores. This indicates that the model is effective at identifying motions related to these topics, likely due to the presence of distinctive keywords and phrases within the text.

Similarly, the class *Further devolution to Scotland* achieved perfect recall, suggesting that the model can reliably detect this category. This may be attributed to the specificity of the language used in motions related to Scottish governance.

However, performance was less consistent for other classes. For example, *More powers for local councils* and *Schools - Greater Autonomy* showed lower recall scores, indicating that the model struggled to correctly identify some instances of these topics. This can be explained by the semantic overlap between policy areas, where similar terminology may be used across different topics, making classification more challenging.

The confusion matrix further highlights these misclassifications, showing that certain classes are occasionally confused with one another. In particular, overlaps between governance-related topics and education-related topics contributed to classification errors.

Overall, while the model demonstrates strong performance for several key classes, its effectiveness varies depending on the distinctiveness of the language associated with each topic. This suggests that further improvements could be achieved by incorporating additional contextual features or more advanced text representation techniques

4c. Task 1 Conclusions.

- The model successfully meets several of the client’s conditions, including achieving strong overall performance and correctly identifying key policy areas such as terrorism and welfare. However, it does not fully meet the target accuracy of 86%, achieving approximately 80%

instead. Despite this, the model demonstrates reliable classification across most categories and provides a useful baseline solution.

- The recommended performance metric for monitoring the model is the **F1-score**, as it provides a balanced measure of both precision and recall. This is particularly important in this task due to variations in class performance and the need to ensure that all policy topics are accurately identified.

5. Task 2: Motion importance classification prototype.

Generally, note that to get full marks in this section some independent research will be needed, on the context of the problem (for example, how parliamentary motions work in the UK) and/or on advanced machine learning techniques.

5a. Ethical discussion.

The use of machine learning to classify the importance of parliamentary motions raises several ethical concerns related to bias, subjectivity, and potential misuse. One of the main challenges in this task is that the concept of “importance” is inherently subjective and depends on individual interpretation, political perspective, and societal context. As a result, the labels used to train the model may reflect personal bias, which can be learned and amplified by the model.

A key risk is that the model may systematically favour certain types of policies or viewpoints, leading to biased prioritisation. For example, motions related to national security or economic policy may be consistently labelled as more important, while others such as local governance or education could be undervalued. This could result in an unfair representation of policy priorities if the system were used in real-world decision-making.

Another concern relates to data limitations. The relatively small size of the labelled dataset increases the risk of overfitting and reduces the model’s ability to generalise. Additionally, the dataset may not fully capture the diversity of parliamentary debates, leading to incomplete or skewed learning.

From a Data Hazards perspective, this task involves issues of **measurement bias** (due to subjective labelling), **representation bias** (limited dataset size and scope), and **aggregation bias** (treating all motions uniformly despite contextual differences). These risks highlight the importance of careful interpretation of model outputs.

To mitigate these concerns, the model should be used only as a decision-support tool rather than a fully automated system. Human oversight is essential to ensure that important nuances are not overlooked and that decisions remain fair and transparent. Furthermore, improving label quality, increasing dataset size, and incorporating multiple annotators could help reduce bias and improve reliability.

5b. Data labelling.

To develop the motion importance classification model, a subset of 100 motions was randomly sampled from the dataset and manually labelled as either *important* or *not important*. The labelling process was based on a combination of topic relevance, perceived societal impact, and the urgency of the issue addressed in each motion.

Motions were generally labelled as *important* if they addressed national-level concerns, such as security, economic policy, or environmental issues, or if they had significant long-term implications. Conversely, motions were labelled as *not important* if they were considered less impactful, more localised in scope, or administrative in nature.

While the topic of the motion provided an initial indication of importance, labels were not assigned solely based on the topic. Instead, the textual content of each motion was reviewed to ensure that decisions reflected the specific context and intent of the motion. This introduced controlled variability in the labelling process and avoided rigid rule-based classification.

The final labelled dataset was balanced, with 50 motions labelled as *important* and 50 as *not important*. This balance ensures that the model is not biased towards a particular class and that performance metrics are meaningful.

However, the labelling process is inherently subjective, and there is uncertainty associated with the assigned labels. Different annotators may interpret the importance of a motion differently depending on their perspective. This introduces potential measurement bias, which may affect model performance.

For example, a motion related to national security was labelled as *important*, while a motion concerning minor administrative changes in local governance was labelled as *not important*. These examples illustrate the criteria used and highlight the subjective nature of the task.

5c. Model building and evaluation.

To develop a prototype model for motion importance classification, a supervised machine learning approach was used. The input feature for this task was the *motion_text*, while the target variable was the manually assigned *motion_priority* label.

The textual data was encoded using TF-IDF with a maximum of 8000 features, removal of English stop words, and inclusion of both unigrams and bi-grams. This representation allows the model to capture both individual words and meaningful phrases that contribute to determining the importance of a motion.

A **Logistic Regression** model was selected for this task due to its simplicity, interpretability, and effectiveness in text classification problems. Given that this is a prototype and the dataset is relatively small, a simple model is appropriate and helps reduce the risk of overfitting.

The model was trained using an 80:20 train-test split with stratification to preserve the class distribution. The final model achieved an accuracy of approximately **60%**, compared to a baseline

accuracy of 50% (reflecting the balanced dataset). This indicates that the model is able to learn meaningful patterns from the data and performs better than random or majority-class prediction.

The classification report shows that the model performs better at identifying *not important* motions, achieving higher recall for this class. In contrast, the recall for *important* motions is lower, suggesting that the model struggles to consistently identify cases that should be prioritised. This imbalance in performance highlights the difficulty of the task and reflects the subjective nature of the labels.

The confusion matrix further supports this observation, showing that several *important* motions are misclassified as *not important*. This may be due to the subtle and context-dependent differences between the two classes, which are not always easily captured through textual features alone.

Overall, the model demonstrates moderate performance and successfully outperforms the baseline. However, the results also indicate that the task is challenging and that further improvements could be achieved through larger datasets, more consistent labelling, or the inclusion of additional contextual features.

5d. Task 2 Conclusions.

You should provide three short bullet points with the following:

- The model is partially successful according to the client's definition of success. It achieves an accuracy of 60%, outperforming the baseline of 50%, which indicates that it has learned meaningful patterns from the data. However, its performance is not strong enough to be considered fully reliable, particularly due to its lower recall for *important* motions.
- The task is only partially well framed given the available data. The concept of "importance" is inherently subjective, and the labels depend heavily on individual judgement. Additionally, the relatively small size of the labelled dataset limits the model's ability to generalise and capture nuanced differences between classes.
- A key improvement would be to increase the size and quality of the labelled dataset, ideally by involving multiple annotators to reduce subjectivity and bias. This would provide more consistent training data and improve the model's ability to generalise, leading to better performance.

6. Self-reflection.

The most challenging part of this coursework was defining and labelling motion importance, as it required subjective judgement and consistency. If starting again, more time would be spent planning the labelling criteria and exploring additional features to improve model performance.

7. References.

Scikit-learn Developers (2024) *TfidfVectorizer*. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html (Accessed:

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8. Declaration of AI Use

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – NO
- To create an outline of the topic – YES
- To explain concepts – YES
- To support my use of language – YES
- To summarise the following articles/resources:

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

- In other ways, as described below:

AI was used to guide the development of machine learning workflows, assist with debugging code, and provide feedback on model evaluation and report structure. All implementation, data labelling, and final decisions were completed independently.